

NIIGATA, Japan

WMO: 476040

Lat: 37.8936N

Lon: 139.0189E

Elev: 6

StdP: 101.26

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	-1.0	-7.7	2.0	1.4	-6.8	2.1	1.8	11.6	3.7	10.8	2.9	4.3	300	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	33.0	25.2	31.7	24.9	30.4	24.5	26.1	30.8	25.6	30.1	25.1	29.4	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	19.9	28.8	24.2	19.2	28.5	23.7	18.5	28.1	81.2	30.9	78.9	30.2	76.7	29.6	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.0	8.7	7.5	-3.4	35.8	1.0	1.2	-4.2	36.7	-4.7	37.4	-5.3	38.0	-6.0	38.9	
			DB												
			WB	-4.5	27.0	0.9	0.6	-5.2	27.4	-5.8	27.8	-6.3	28.1	-7.0	28.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	2.9	3.1	6.2	11.6	17.3	21.2	25.3	27.0	22.9	17.0	10.7	5.6	(6)
		DBStd	8.79	2.11	2.45	2.92	3.42	3.05	2.30	2.58	2.41	3.00	2.85	3.09	2.66	
		HDD10.0	723	221	192	123	21	0	0	0	0	0	28	139		
		HDD18.3	2229	479	426	376	202	57	4	0	1	61	228	396		
		CDD10.0	2296	0	1	6	70	225	336	476	526	388	216	51	2	
		CDD18.3	757	0	0	0	2	23	89	217	268	139	18	0	0	
		CDH23.3	5707	0	0	0	9	96	293	1711	2702	857	39	0	0	
		CDH26.7	1767	0	0	0	1	11	39	530	963	218	4	0	0	
(14)	Wind	WSAvg	3.3	4.2	3.9	3.7	3.5	3.2	2.8	2.8	2.8	2.9	3.0	3.4	4.0	(14)
(15)	Precipitation	PrecAvg	1817	187	126	107	97	98	116	191	145	164	155	199	231	(15)
(16)		PrecMax	2397	360	262	182	150	215	378	443	616	335	285	416	444	(16)
(17)		PrecMin	1467	106	57	42	34	35	38	15	4	26	16	85	98	(17)
(18)		PrecStd	238	55	40	29	29	44	68	111	114	74	51	71	77	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	11.0	13.5	18.7	24.6	28.1	30.1	34.3	35.1	33.3	26.5	20.9	15.2		(19)
(20)		0.4% MCWB	7.5	8.2	10.7	14.6	18.3	21.6	25.1	25.4	24.6	19.7	14.8	11.2		(20)
(21)		2% DB	8.8	10.5	15.6	21.6	25.8	27.6	32.5	33.4	30.9	24.2	18.5	12.8		(21)
(22)		2% MCWB	5.9	7.0	9.4	13.4	17.0	20.6	25.0	25.4	24.0	18.8	13.8	9.8		(22)
(23)		5% DB	7.4	8.7	13.0	19.3	24.1	26.2	31.1	32.2	29.1	22.8	17.0	11.3		(23)
(24)		5% MCWB	4.7	5.4	8.4	12.6	16.4	20.2	24.8	25.3	23.2	18.0	13.2	8.4		(24)
(25)		10% DB	6.3	7.2	11.1	17.2	22.3	24.9	29.6	31.0	27.5	21.6	15.6	9.8		(25)
(26)		10% MCWB	3.8	4.3	7.5	11.7	16.0	19.7	24.3	25.1	22.5	17.1	12.3	7.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	8.5	9.4	11.9	16.6	19.9	23.3	26.5	27.1	25.8	21.6	16.3	12.3		(27)
(28)		0.4% MCDB	10.2	12.3	16.2	20.2	24.3	26.5	31.7	31.9	30.7	24.6	18.4	14.3		(28)
(29)		2% WB	6.3	7.3	10.3	14.6	18.6	22.2	25.7	26.4	24.6	20.1	15.0	10.1		(29)
(30)		2% MCDB	8.2	9.9	13.9	19.4	23.3	25.3	30.7	31.0	29.4	22.9	17.6	12.7		(30)
(31)		5% WB	5.2	5.8	9.1	13.5	17.7	21.5	25.2	26.0	23.8	18.9	13.7	8.6		(31)
(32)		5% MCDB	7.0	8.3	12.3	17.7	22.2	24.7	29.7	30.3	28.0	21.8	16.2	10.8		(32)
(33)		10% WB	4.1	4.5	7.9	12.4	16.8	20.7	24.6	25.5	23.1	17.7	12.6	7.4		(33)
(34)		10% MCDB	6.0	6.9	11.0	16.4	21.0	24.0	28.8	29.8	26.8	20.8	15.3	9.5		(34)
(35)	Mean Daily Temperature Range	MDBR	4.6	5.4	6.9	7.8	7.7	6.1	5.8	6.3	6.3	6.8	6.5	5.3		(35)
(36)		5% DB MCDBR	6.7	8.3	11.0	11.6	10.4	8.0	7.9	7.8	8.0	8.0	8.8	7.6		(36)
(37)		5% DB MCWBR	5.1	6.1	7.1	5.9	4.7	3.6	3.2	3.2	3.7	4.4	5.8	5.9		(37)
(38)		5% WB MCDBR	6.5	8.1	9.3	9.1	8.6	6.5	7.1	7.0	6.3	7.5	7.5	7.1		(38)
(39)		5% WB MCWBR	5.5	6.5	6.8	5.3	4.7	3.3	3.0	3.2	3.4	4.2	5.9	6.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.420	0.455	0.517	0.550	0.556	0.561	0.569	0.545	0.484	0.448	0.432	0.426		(40)
(41)		taud	1.992	1.910	1.832	1.814	1.864	1.932	1.972	2.042	2.169	2.154	2.100	2.017		(41)
(42)		Ebn at Noon	712	743	739	745	750	745	735	742	767	750	701	671		(42)
(43)		Edh at Noon	132	162	193	208	202	189	180	164	137	126	118	121		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.40	2.22	3.42	4.53	5.22	5.16	4.69	4.91	3.89	2.91	1.83	1.26		(44)
(45)		RadStd	0.14	0.24	0.30	0.42	0.54	0.40	0.67	0.53	0.30	0.22	0.15	0.15		(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.29	N/A	N/A	N/A	N/A

Nomenclature: See separate page